BAOMING ZHANG

Ph.D. Candidate in Computer Science | Computer Vision & Robust Machine Learning
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Research: Vision-Language & Foundation Models | Test-Time Adaptation | Open-Vocabulary & 3D Perception | Generative AI

EDUCATION

The University of Texas at Dallas (UT Dallas) | Richardson, TX **2023 - May 2028 (expected)**
Ph.D. in Computer Science (GPA: 3.8/4.0) | M.S. in Computer Science, Dec 2025 (GPA: 3.8/4.0)
Chongqing University (CQU) | Chongqing, China **Jul 2021**
M.S. in Control Science and Engineering (GPA: 3.7/4.0)

RESEARCH EXPERIENCE

Graduate Researcher, UT Dallas **Sep 2023 - Present**

- **Grounding DINO TTA (Ongoing):** Developing a noise-aware test-time adaptation framework for robust open-vocabulary object detection under environmental corruptions, using instance-level DINOv2 consistency learning as the adaptation objective.
- **LH3D - Data-Efficient 3D Perception (CVPR 2026):** Co-developed a three-stage learnability-driven active-learning framework for roadside monocular 3D detection; used 32,000 labeled objects (12.98% of the training pool) and outperformed PPAL by up to 2.19 average 3D AP points on DAIR-V2X-I; evaluated on DAIR-V2X-I with three detector backbones and on Rope3D.
- **SafeFix - Foundation-Model Reliability (TMLR 2026):** Co-developed an automated model-repair pipeline powered by Stable Diffusion/ControlNet and Qwen2.5-VL/LLaVA; improved test accuracy by up to 2.89 points on CelebA and increased ImageNet10 ResNet-18 accuracy from 71.73% to 74.88%; evaluated with ResNet, ViT, and CLIP without introducing new bugs.
- **BATCLIP - Vision-Language Model Adaptation (ICCV 2025):** Co-developed a bimodal online TTA method that updated only ~0.044% of CLIP parameters, improved CIFAR-10C accuracy by 6.48 points over SAR, and generalized across CIFAR-10C/100C, ImageNet-C, and domain shifts.
- **AVROBUSTBENCH - Multimodal Robustness (NeurIPS 2025 D&B):** Benchmarked supervised, self-supervised, and online TTA models across four audio-visual datasets, each with 15 co-occurring bimodal corruptions at five severity levels (75 corruption conditions), exposing robustness degradation under bimodal distribution shifts.
- **PALM - Continual Test-Time Adaptation (AAAI 2025 Oral):** Co-designed an adaptive layer-wise CTTA method that reduced mean error vs. LAW by 1.7/1.6/0.9 points on CIFAR-10C/100C/ImageNet-C while updating only 4.34%/6.65%/17.21% of parameters per task.

Research Assistant, University of Missouri-Columbia **Aug 2022 - Aug 2023**

- Built segmentation-conditioned residual diffusion models for anatomically faithful abdominal CT synthesis; validated image fidelity with SSIM/FID and downstream segmentation performance.

SELECTED PUBLICATIONS

[CVPR 2026] R. Mao*, **B. Zhang***, N. Ruozzi, Y. Guo. *Learnability-Driven Submodular Optimization for Active Roadside 3D Detection.*

[TMLR 2026] O. Xu, **B. Zhang**, R. Mao, Y. Guo. *SafeFix: Targeted Model Repair via Controlled Image Generation.*

[ISBI 2026 Oral] D. He*, **B. Zhang***, X. Wang*, et al. *PIVM: Diffusion-Based Prior-Integrated Variation Modeling for Anatomically Precise Abdominal CT Synthesis.*

[NeurIPS 2025 D&B] S. K. Maharana, S. S. Kushwaha, **B. Zhang**, et al. *AVROBUSTBENCH: Benchmarking the Robustness of Audio-Visual Recognition Models at Test-Time.*

[ICCV 2025] S. K. Maharana, **B. Zhang**, L. Karlinsky, R. Feris, Y. Guo. *BATCLIP: Bimodal Online Test-Time Adaptation for CLIP.*

[AAAI 2025 Oral] S. K. Maharana, **B. Zhang**, Y. Guo. *PALM: Pushing Adaptive Learning Rate Mechanisms for Continual Test-Time Adaptation.*

** Equal contribution*

TECHNICAL SKILLS

Research: Vision-language and foundation models, open-vocabulary detection, test-time adaptation, model reliability/repair, active learning, 3D perception
Models: CLIP, DINOv2, SAM, Grounding DINO, Stable Diffusion, ControlNet, Qwen2.5-VL, LLaVA
Programming: Python, C++, MATLAB
Frameworks & Tools: PyTorch, OpenCV, Hugging Face, PyTorch Lightning, TensorFlow, scikit-learn, Docker, Git, LaTeX
Open Source: LH3D (CVPR 2026) | PIVM (ISBI 2026 Oral)

SERVICE

Reviewer: ECCV 2024/2026; CVPR 2025/2026; ISBI 2026; BMVC 2026; ICIP 2026; NeurIPS 2026; Medical Physics